

Last login: Mon Jun 22 13:46:05 on ttys001
adityadave@Adityas-MacBook-Air-3 ~ % cd /Users/adityadave/PE/PE-Daywise-Work-Tracker/19-JUNE/Q2\&Q21
adityadave@Adityas-MacBook-Air-3 Q2&Q21 % python3 script.py
Running 1/10...
Running 2/10...
Running 3/10...
Running 4/10...
Running 5/10...
Running 6/10...
Running 7/10...
Running 8/10...
Running 9/10...
Running 10/10...

=====

10TH RUN OUTPUT

=====

SET
WARNING: Log 1: Planner Input:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:ctelist <>
:rtable (
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname supplier
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:relid 17399
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname l1
:colnames <>
}
:eref
{ALIAS
:aliasname l1
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_qua
ntity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines
tatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "
l_shipmode" "l_comment")
}
:rtekind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber"
"l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnfl
ag" "l_linestatus" "l_shipdate" "l_commitdate" "l_receiptdate"
"l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 17440
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17440
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0

```

:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
  {RANGETBENTRY
    :alias
      {ALIAS
        :aliasname o
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname o
        :colnames ("o_orderkey" "o_custkey" "o_orderstatus"
          "o_totalprice" "o_orderdate" "o_orderpriority" "o_
            clerk" "o_shippriority" "o_comment")
      }
    :rtekind 0
    :relid 17428
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 17428
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
  }
:quals
  {BOOLEXP
    :boolop and
    :args (
      {OEXPR
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location -1
          }
          {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 1
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location -1
          }
        )
      }
      {location -1
      }
    )
  }
  {SCALARARRAYOEXPR
    :opno 1054
    :opfuncid 1048
    :hashfuncid 0
    :negfuncid 0
    :useOR true
    :inputcollid 100
    :args (
      {VAR
        :varno 1
        :varattno 6
        :vartype 1042
        :vartypmod 19
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 6
        :location -1
      }
    )
  }
  {ARRAYEXPR
    :array_typeid 1014
    :array_collid 100
    :element_typeid 1042
    :elements (
      {CONST
        :consttype 1042
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 12 [ 48 0 0 0 49 45 85 82 71 6
          9 78 84 ]
      }
      {CONST
        :consttype 1042
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 10 [ 40 0 0 0 50 45 72 73 71 7
          2 ]
      }
    )
  }
)
:multidims false
:list_start -1

```

```

        :list_end -1
        :location -1
    }
    )
    :location -1
    }
    )
    :location -1
    }
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:ReturningOldAlias <>
:ReturningNewAlias <>
:ReturningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
:windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty"
"ps_supplycost" "ps_comment")
}
:relkind 0
:relid 17409
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17409
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcoid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varstype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1

```

```

        :location -1
      }
      {VAR
        :varno 1
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 1
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 2
        :location -1
      }
    }
    :location -1
  }
  {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 2
        :location -1
      }
      {VAR
        :varno 1
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 1
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 3
        :location -1
      }
    )
    :location -1
  }
  {OPEXPR
    :opno 521
    :opfuncid 147
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 3
        :location -1
      }
      {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 0 0 0 0 0 0 0 0 ]
      }
    )
    :location -1
  }
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location -1
      :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>

```

```

:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname l_orderkey
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname l_partkey
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname l_suppkey
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname l_linenumber
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname l_quantity
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname l_extendedprice
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
}

```

```

:resno 7
:resname l_discount
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location -1
}
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location -1
}
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location -1
}
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)

```

```

:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location -1
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:varatype 1042
:vartypmod 14
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location -1
}
:resno 15
:resname l_shipmode
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:varatype 1043
:vartypmod 48
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location -1
}
:resno 16
:resname l_comment
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 16
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returninglist <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier false
:reldid 17466
:inh false
:relkind v
:rellockmode 1
:perminfoindex 2
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:relkind 0
:reldid 17428
:inh true
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:relkind 0
:reldid 17376
:inh true
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("s_name")
}
:relkind 9
:groupexprs (
{VAR
:varno 1
:varattno 2
:varatype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 51

```

```

    }
    :lateral false
    :inFromCl false
    :securityQuals <>
  }
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17399
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17466
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10 19 20)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17428
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17376
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
)
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 183
}
{VAR
:varno 2
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location 195
}
)
}
:location 193
)
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 220
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 233
}
)
}
:location 231
)
{OPEXP
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR

```

```

:varno 3
:varattno 3
:varatype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 3
:location 259
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 275
:constvalue 5 [ 20 0 0 0 70 ]
}
}
:location 273
}
{OPEXPR
:opno 1097
:opfuncid 1089
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 13
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 13
:location 291
}
{VAR
:varno 2
:varattno 12
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 12
:location 310
}
}
:location 308
}
{SUBLINK
:sublinktype 0
:sublinkid 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname 12
:colnames <>
}
:eref
{ALIAS
:aliasname 12
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
"l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
t")
}
:rtekind 0
:relid 17440
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17440
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
23)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXPR
:boolop and
:args (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1

```

```

        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location 427
    }
    {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location 443
    }
    }
    :location 441
    }
    {OPEXPR
    :opno 518
    :opfuncid 144
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location 479
    }
    {VAR
    :varno 2
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 3
    :location 495
    }
    )
    :location 492
    }
    }
    :location 475
    }
    }
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 370
}
}
:resno 1
:resname l_orderkey
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 370
}
}
:resno 2
:resname l_partkey
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 370
}
}
:resno 3
:resname l_supkey
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0

```

```

:varnosyn 1
:varattnosyn 4
:location 370
}
:resno 4
:resname 1_linenummer
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:varatype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 370
}
:resno 5
:resname 1_quantity
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:varatype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 370
}
:resno 6
:resname 1_extendedprice
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:varatype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 370
}
:resno 7
:resname 1_discount
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:varatype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 370
}
:resno 8
:resname 1_tax
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:varatype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 370
}
:resno 9
:resname 1_returnflag
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:varatype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 370
}
:resno 10
:resname 1_linestatus
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11

```

```

:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 370
}
}
:resno 11
:resname 1_shipdate
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 370
}
}
:resno 12
:resname 1_commitdate
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 370
}
}
:resno 13
:resname 1_receiptdate
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location 370
}
}
:resno 14
:resname 1_shipinstruct
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 1042
:vartypmod 14
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location 370
}
}
:resno 15
:resname 1_shipmode
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 1043
:vartypmod 48
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location 370
}
}
:resno 16
:resname 1_comment
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 16
:resjunk false
}
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0

```

```

:stmt_len 0
}
:location 338
}
{BOOLEXP
:boolop not
:args (
{SUBLINK
:sublinkType 0
:sublinkid 0
:texexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname l3
:colnames <>
}
:eref
{ALIAS
:aliasname l3
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_lin
number" "l_quantity" "l_extendedprice" "l_discount" "
l_tax" "l_returnflag" "l_linestatus" "l_shipdate" "l_c
ommitdate" "l_receiptdate" "l_shipinstruct" "l_shipmod
e" "l_comment")
}
:rtekind 0
:relid 17440
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17440
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21
22 23)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varstype 23
:varbypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 623
}
{VAR
:varno 2
:varattno 1
:varstype 23
:varbypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 639
}
)
}
:location 637
}
{OPEXP
:opno 518
:opfuncid 144
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:varstype 23
:varbypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 675
}
{VAR
:varno 2
:varattno 3
:varstype 23
:varbypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1

```

```

        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 3
        :location 691
    }
}
:location 688
}
{OPEXPR
:opno 1897
:opfuncid 1089
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 726
}
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 745
}
)
:location 743
}
):location 671
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 566
}
:resno 1
:resname l_orderkey
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 566
}
:resno 2
:resname l_partkey
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 566
}
:resno 3
:resname l_suppkey
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 566
}
:resno 4
:resname l_linenummer
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046

```

```

:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 566
}
:resno 5
:resname l_quantity
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 566
}
}
:resno 6
:resname l_extendedprice
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 566
}
}
:resno 7
:resname l_discount
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 566
}
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 566
}
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 566
}
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 566
}
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 11
:resjunk false
}
{TARGETENTRY

```

```

:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 566
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 566
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location 566
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 1042
:vartypmod 14
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location 566
}
:resno 15
:resname l_shipmode
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 1043
:vartypmod 48
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location 566
}
:resno 16
:resname l_comment
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 16
:resjunk false
}
}
:location 534
}
}
:location 530
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 4

```

```

        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 4
        :location 783
    }
    {VAR
        :varno 4
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 1
        :location 797
    }
}
:location 795
}
{OPEXPR
:opno 1054
:opfncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
    {VAR
        :varno 4
        :varattno 2
        :vartype 1042
        :vartypmod 29
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 2
        :location 821
    }
    {CONST
        :consttype 1042
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location 830
        :constvalue 13 [ 52 0 0 0 65 82 71 69 78 84 73 78 65 ]
    }
)
:location 828
}
)
:location 216
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
        :expr
        {VAR
            :varno 5
            :varattno 1
            :vartype 1042
            :vartypmod 29
            :varcollid 100
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 5
            :varattnosyn 1
            :location -1
        }
        :resno 1
        :resname s_name
        :ressortgroupref 2
        :resorigtbl 17399
        :resorigcol 2
        :resjunk false
    }
    {TARGETENTRY
        :expr
        {AGGREF
            :aggfnoid 2803
            :aggtype 20
            :aggcollid 0
            :inputcollid 0
            :aggtranstype 0
            :aggargtypes <>
            :aggdirectargs <>
            :args <>
            :aggorder <>
            :aggdistinct <>
            :aggfilter <>
            :aggstar true
            :aggvariadic false
            :aggkind n
            :aggpresorted false
            :agglevelsup 0
            :aggsplit 0
            :aggno -1
            :aggtransno -1
            :location 67
        }
        :resno 2
        :resname numwait
        :ressortgroupref 1
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
    {SORTGROUPCLAUSE
        :tleSortGroupRef 2
        :eqop 1054
        :sortop 1058
        :reverse_sort false
        :nulls_first false
        :hashable true
    }
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
    {SORTGROUPCLAUSE
        :tleSortGroupRef 1
        :eqop 410
        :sortop 413
        :reverse_sort true
    }
)

```

```

:nulls_first true
:hashable true
}
{SORTGROUPCLAUSE
:tleSortGroupRef 2
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: Log 2 : After pull up At starting of Path Generation:

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname supplier
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:relid 17399
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname l1
:colnames <>
}
:eref
{ALIAS
:aliasname l1
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_qua
ntity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines
tatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "
l_shipmode" "l_comment")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 17466
:inh false
:relkind v
:rellockmode 1
:perminfoindex 2
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o
orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 17428
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 17376
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("s_name")
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 2
:varstype 1042
:varitypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 51
}
)
}
)

```

```

:lateral false
:inFromCl false
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname l2
:colnames <>
}
}
:eref
{ALIAS
:aliasname l2
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
}
:rtekind 0
:relid 17440
:inh false
:relkind r
:rellockmode 1
:perminfoindex 5
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname l3
:colnames <>
}
}
:eref
{ALIAS
:aliasname l3
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
}
:rtekind 0
:relid 17440
:inh false
:relkind r
:rellockmode 1
:perminfoindex 6
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
}
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
}
:rtekind 0
:relid 17440
:inh false
:relkind r
:rellockmode 1
:perminfoindex 7
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname o
:colnames <>
}
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
}
:rtekind 0
:relid 17428
:inh false
:relkind r
:rellockmode 1
:perminfoindex 8
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "ps_comment")
}
}
:rtekind 0
:relid 17409
:inh false
:relkind r
:rellockmode 1
:perminfoindex 9
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17399
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17466
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10 19 20)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17428
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
}
)

```

```

:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17376
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17440
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17440
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17440
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17428
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17409
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
}
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 5
:isNatural false
:larg
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{JOINEXPR
:jointype 4
:isNatural false
:larg
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 8
}
)
:quals <>
}
:rarg
{RANGETBLREF
:rtindex 9
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 9
:varattno 1
:vartype 23
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 9
:varattnosyn 1
:location -1
}
{VAR
:varno 8
:varattno 1
:vartype 23
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 8
:varattnosyn 1
:location -1
}
)
:location -1
}
{SCALARARRAYOPEXPR
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
{VAR
:varno 9
:varattno 6
:vartype 1042
:varitypmod 19
:varcollid 100
:varnullingrels (b)

```

```

:varlevelsup 0
:varreturningtype 0
:varnosyn 9
:varattnosyn 6
:location -1
}
{CONST
:consttype 1014
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 48 [ -64 0 0 0 1 0 0 0 0 0 0 0 0 18 4 0 0
2 0 0 0 1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 4
0 0 0 0 50 45 72 73 71 72 0 0 ]
}
)
:location -1
}
)
:alias <>
:rtindex 0
}
:rarg
{RANGETBLREF
:rtindex 10
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 10
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 10
:varattnosyn 1
:location -1
}
{VAR
:varno 8
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 8
:varattnosyn 2
:location -1
}
)
:location -1
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 10
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 10
:varattnosyn 2
:location -1
}
{VAR
:varno 8
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 8
:varattnosyn 3
:location -1
}
)
:location -1
}
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 10
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 10
:varattnosyn 3
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:alias <>
:rtindex 0
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF

```

```

:rtindex 4
}
)
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 183
}
{VAR
:varno 8
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 8
:varattnosyn 3
:location -1
}
)
:location 193
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 220
}
{VAR
:varno 8
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 8
:varattnosyn 1
:location -1
}
)
:location 231
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 3
:varattno 3
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 3
:location 259
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 275
:constvalue 5 [ 20 0 0 0 70 ]
}
)
:location 273
}
{OPEXPR
:opno 1097
:opfuncid 1089
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 8
:varattno 13
:vartype 1002
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 8
:varattnosyn 13
:location -1
}
{VAR
:varno 8
:varattno 12
:vartype 1002
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 8
:varattnosyn 12
:location -1
}
)
:location 308

```

```

    }
    {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 1
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 4
    :location 783
    }
    {VAR
    :varno 4
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 1
    :location 797
    }
    )
    :location 795
    }
    {OPEXPR
    :opno 1054
    :opfuncid 1048
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 100
    :args (
    {VAR
    :varno 4
    :varattno 2
    :vartype 1042
    :vartypmod 29
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 2
    :location 821
    }
    {CONST
    :consttype 1042
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location 830
    :constvalue 13 [ 52 0 0 0 65 82 71 69 78 84 73 78 65 ]
    }
    )
    :location 828
    }
    }
    :rarg
    {RANGETBLREF
    :rtindex 6
    }
    :usingClause <>
    :join_using_alias <>
    :quals (
    {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 6
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 6
    :varattnosyn 1
    :location 427
    }
    {VAR
    :varno 8
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 8
    :varattnosyn 1
    :location -1
    }
    )
    :location 441
    }
    {OPEXPR
    :opno 518
    :opfuncid 144
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 6
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 6
    :varattnosyn 3
    :location 479
    }
    {VAR
    :varno 8
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0

```

```

        :varnosyn 8
        :varattnosyn 3
        :location -1
    }
}
:location 492
}
)
:alias <>
:rtindex 0
}
:rarg
{RANGETBLREF
:rtindex 7
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 1
:location 623
}
{VAR
:varno 8
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 8
:varattnosyn 1
:location -1
}
)
:location 637
}
{OPEXPR
:opno 518
:opfuncid 144
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 3
:location 675
}
{VAR
:varno 8
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 8
:varattnosyn 3
:location -1
}
)
:location 688
}
{OPEXPR
:opno 1097
:opfuncid 1089
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 13
:location 726
}
{VAR
:varno 7
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 12
:location 745
}
)
:location 743
}
)
:alias <>
:rtindex 0
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100

```

[illegible]

```
WARNING: is_parallel_safe: INPUT qual = <> | parallel_safe = true | reason = PROPAGALABLE_SAFE
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPAGALABLE_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 6 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattsyns 1 :location -1) (VAR :varno 6 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattsyns 3 :location -1))
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 6 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattsyns 1 :location -1) (VAR :varno 6 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattsyns 3 :location -1)) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OEXPR :opno 1897 :opfncid 1889 :opsresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 7 :varattno 13 :vartype 1
082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varattsyns 13 :location -1) (VAR :varno 7 :varattno 12 :vartype 1082 :vartypmod -1 :varcollid 0 :varnu
llingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varattsyns 12 :location -1)) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :lea
kproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 7) :required_relds (b 7) :incompatible_relds (b) :outer_relds (b 1 3 4 6 8 9 10) :left_relds (b 7) :right_relds (b
7) :orclause <> :rinfo_serial 9 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selc -1 :outer_selc -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_b
ucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0)) | parallel_safe = true | reason = PROPAGALABLE_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause (OEXPR :opno 1897 :opfncid 1889 :opsresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 7 :varattno 13 :vartype
1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varattsyns 13 :location -1) (VAR :varno 7 :varattno 12 :vartype 1082 :vartypmod -1 :varcollid 0 :varn
ullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varattsyns 12 :location -1)) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :le
akproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 7) :required_relds (b 7) :incompatible_relds (b) :outer_relds (b 1 3 4 6 8 9 10) :left_relds (b 7) :right_relds (b
7) :orclause <> :rinfo_serial 9 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selc -1 :outer_selc -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_b
ucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0)) | parallel_safe = true | reason = PROPAGALABLE_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 7 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varattsyns 1 :location -1)
(VAR :varno 7 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varattsyns 3 :location -1))
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 7 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varattsyns 1 :location -1)
(VAR :varno 7 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varattsyns 3 :location -1)) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OEXPR :opno 1897 :opfncid 1889 :opsresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 8 :varattno 13 :vartype 1
082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 8 :varattsyns 13 :location -1) (VAR :varno 8 :varattno 12 :vartype 1082 :vartypmod -1 :varcollid 0 :varnu
llingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 8 :varattsyns 12 :location -1)) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :le
akproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 8) :required_relds (b 8) :incompatible_relds (b) :outer_relds (b 8) :left_relds (b 8) :right_relds (b 8) :orclause <> :r
info_serial 9 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selc -1 :outer_selc -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :ri
ght_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0))
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause (OEXPR :opno 1897 :opfncid 1889 :opsresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 8 :varattno 13 :vartype
1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 8 :varattsyns 13 :location -1) (VAR :varno 8 :varattno 12 :vartype 1082 :vartypmod -1 :varcollid 0 :varn
ullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 8 :varattsyns 12 :location -1)) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :le
akproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 8) :required_relds (b 8) :incompatible_relds (b) :outer_relds (b 8) :left_relds (b 8) :right_relds (b 8) :orclause <> :r
info_serial 9 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selc -1 :outer_selc -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :ri
ght_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0)) | parallel_safe = true | reason = PROPAGALABLE_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 8 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 8 :varattsyns 1 :location -1)
(VAR :varno 8 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 8 :varattsyns 2 :location -1)) (VAR :varno 8 :varattno 3 :vartype 23 :
vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 8 :varattsyns 3 :location -1))
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (SCALARARRAYOEXPR :opno 1854 :opfncid 1848 :hashfuncid 0 :negfncid 0 :useur true :inputcollid 100 :args ((VAR :varno 9 :varattno 6 :va
rtypmod 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 9 :varattsyns 6 :location -1) (CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen
-1 :constbval false :constnull false :location -1) :constval 48 [-64 0 0 1 0 0 0 0 0 0 18 4 0 0 2 0 0 1 0 0 0 48 0 0 49 85 82 71 69 78 84 0 0 0 50 45 72 73 71 72 0 0 ]]) :location -1)
:is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 9) :required_relds (b 9)
:incompatible_relds (b) :outer_relds (b) :left_relds (b) :orclause <> :rinfo_serial 9 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selc -1 :outer_selc -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0)) | paral
el_safe = true | reason = PROPAGALABLE_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 9 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 9 :varattsyns 1 :location -1)
(VAR :varno 9 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 9 :varattsyns 2 :location -1)) (VAR :varno 9 :varattno 3 :vartype 23 :
vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 9 :varattsyns 3 :location -1))
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 9 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 9 :varattsyns 1 :location -1)
| parallel_safe = true | reason = PROPAGALABLE_SAFE
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OEXPR :opno 521 :opfncid 147 :opsresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 10 :varattno 3 :vartype 23 :
vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 10 :varattsyns 3 :location -1) (CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbval t
rue :location -1) :constval 4 [0 0 0 0 0 0 0 0 ]]) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_vol
atile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 10) :required_relds (b 10) :incompatible_relds (b) :outer_relds (b 10) :left_relds (b 10) :right_relds (b 10) :orclause <> :rinfo_ser
ial 5 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selc -1 :outer_selc -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :l
eft_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0))
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause (OEXPR :opno 521 :opfncid 147 :opsresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 10 :varatt
```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```
WARNING: iss_parallel_safe: OUTPUT qual = ((RESTRINFO clause @OPEXPR :opno 518 :opfuncid 144 :opsresultype 16 :opretset false :opcollid 0 :inputcollid 0 :args {fVAR :varno 7 :varattno 3 :vartyp 23 :vartymod -1 :varcollid 0 :varnullingrels {b :varlevelslop 0 :varreturntuninginfo 0 :varsynon 7 :varattsynon 3 :location -1} fVAR :varno 8 :varattno 3 :vartyp 23 :vartymod -1 :varcollid 0 :varnullingrels {b :varlevelslop 0 :varreturntuninginfo 0 :varsynon 8 :varattsynon 3 :location -1} :location -1}) :is_pushed_down false :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 2 :clause_relds {b 7 8} :required_relds {b 7 8} :incompatible_relds {b } :outer_relds {b 1 3 4 6 8 9 10} :left_relds {b 7} :right_relds {b 8} :relocate < :rinfo serial 15 :label cost.startup 0 :total_cost.per_tuple 0.0025 :norm_selcost 0.99949679733582758 :outer_select 1 :mergeofamilies < :left_em <> :right_em <> :outer_is_left false :hashjoinorder array [left_bucketsize := left_bucketsize :eval_freq_per_tuple mcfreq := left_hashseqoperator %6] RESTRINFO clause @OPEXPR :opno 76 :opfuncid 95 :opsresultype 16 :opretset false :opcollid 0 :inputcollid 0 :args {fVAR :varno 7 :varattno 3 :vartyp 23 :vartymod -1 :varcollid 0 :varnullingrels {b :varlevelslop 0 :varreturntuninginfo 0 :varsynon 7 :varattsynon 3 :location -1} fVAR :varno 8 :varattno 3 :vartyp 23 :vartymod -1 :varcollid 0 :varnullingrels {b :varlevelslop 0 :varreturntuninginfo 0 :varsynon 8 :varattsynon 3 :location -1}})
```

```
--(VAR :varno 8 :varattino 1 :vartype 23 :is_volatile -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyns 8 :varattosyn 1 :location -1) :location -1 ;:is_pushed_down true;can_join true;pseudocountant false;has_clone false;is_clone false;leakproof false;has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 7 8) :required_relds (b 7 8) :incompatible_rel_reids (b) :outer_relds ((b 1 3 4 6 8 9 10) :left_relds (b 7) :right_relds (b 8)) :orclause <> :rinfo_serial 14 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selc 0.8171666666666667 :outer_select 1 :mergeofamilies (o 1976) :left_em <> :right_em <> :router_is_left true :hashjoinoperator %6 :left_bucketsize 1.2237355752169072e-05 :right_bucketsize 1.2237355752169072e-05 :left_mvfreq 0 :right_mvfreq 0 :left_hashselector %6 :right_hashselector %6);| parallel_safe = true | reason = PROPARALLEL_SAFE  
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 8 :varattino 1 :vartype 2 :vartype 1842 :varcollid 100 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 1 :varattosyn 2 :location -1} :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 1 :varattosyn 1 :location -1}) (VAR :varno 8 :varattino 1 :vartype 2 :vartype 23 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 8 :varattosyn 1 :location -1})  
(b) :varlevelsups 0 :varreturningtype 0 :varnosyn 8 :varattosyn 1 :location -1) :location -1 ;:is_pushed_down true;can_join true;pseudocountant false;has_clone false;is_clone false;leakproof false;has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 3 9) :required_relds (b 3 9) :incompatible_relds (b) :outer_relds ((b 3) :right_relds (b 9)) :orclause <> :rinfo_serial 25 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selc 0.3333333333333333e-06 :outer_select 0.4022 :mergeofamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 3 :varattino 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 3 :varattosyn 1 :location -1} :em_relds (b 3)}) :em_is_const false :em_is_child false :em_data_type 23 :em_jdomain (JOINDOMAIN :jd_relds ((b 1 3 4 6 7 8 9 10))) :right_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 3 :varattino 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 3 :varattosyn 1 :location -1}) :em_relds ((b 3 4 6 7 8 9 10))) :out_rer_ids ((b 3 4 6 7 8 9 10))) :router_is_left true :hashjoinoperator %6 :left_bucketsize 0.859651525297822915e-06 :right_bucketsize 0.82875670445881e-06 :left_mvfreq 0 :right_mvfreq 0 :left_hashselector %6 :right_hashselector %6);| parallel_safe = true | reason = PROPARALLEL_SAFE  
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause (OPEXPR :opnpo 65 :opsresultype 16 :opretset false :opolcid 0 :inputcollid 0 :args ((VAR :varno 3 :varattino 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 3 :varattosyn 1 :location -1}) (VAR :varno 3 :varattino 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 3 :varattosyn 1 :location -1})) :location -1} :is_pushed_down true;can_join true;pseudocountant false;has_clone false;is_clone false;leakproof false;has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 3 9) :required_relds (b 3 9) :incompatible_relds (b) :outer_relds ((b 3) :right_relds (b 9)) :orclause <> :rinfo_serial 25 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selc 0.3333333333333333e-06 :outer_select 0.4022 :mergeofamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 3 :varattino 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 3 :varattosyn 1 :location -1}) :em_relds ((b 3 4 6 7 8 9 10))) :right_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 3 :varattino 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 3 :varattosyn 1 :location -1}) :em_relds ((b 3 4 6 7 8 9 10))) :out_rer_ids ((b 3 4 6 7 8 9 10))) :router_is_left true :hashjoinoperator %6 :left_bucketsize 0.859651525297822915e-06 :right_bucketsize 0.82875670445881e-06 :left_mvfreq 0 :right_mvfreq 0 :left_hashselector %6 :right_hashselector %6);| parallel_safe = true | reason = PROPARALLEL_SAFE  
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 8 :varattino 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 8 :varattosyn 1 :location -1}) (VAR :varno 8 :varattino 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 8 :varattosyn 3 :location -1}) (VAR :varno 8 :varattino 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 8 :varattosyn 1 :location -1})  
(b) :varlevelsups 0 :varreturningtype 0 :varnosyn 8 :varattosyn 3 :location -1) :location -1 ;:is_pushed_down true;can_join true;pseudocountant false;has_clone false;is_clone false;leakproof false;has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 3 9) :required_relds (b 3 9) :incompatible_relds (b) :outer_relds ((b 3) :right_relds (b 9)) :orclause <> :rinfo_serial 25 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selc 0.3333333333333333e-06 :outer_select 0.4022 :mergeofamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 3 :varattino 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 3 :varattosyn 1 :location -1}) :em_relds ((b 3 4 6 7 8 9 10))) :right_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 3 :varattino 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 3 :varattosyn 1 :location -1}) :em_relds ((b 3 4 6 7 8 9 10))) :out_rer_ids ((b 3 4 6 7 8 9 10))) :router_is_left true :hashjoinoperator %6 :left_bucketsize 0.859651525297822915e-06 :right_bucketsize 0.82875670445881e-06 :left_mvfreq 0 :right_mvfreq 0 :left_hashselector %6 :right_hashselector %6);| parallel_safe = true | reason = PROPARALLEL_SAFE  
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 8 :varattino 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 8 :varattosyn 1 :location -1}) (VAR :varno 8 :varattino 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 8 :varattosyn 3 :location -1}) (VAR :varno 8 :varattino 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 8 :varattosyn 1 :location -1})  
(b) :varlevelsups 0 :varreturningtype 0 :varnosyn 8 :varattosyn 3 :location -1) :location -1 ;:is_pushed_down true;can_join true;pseudocountant false;has_clone false;is_clone false;leakproof false;has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 6 8) :required_relds (b 6 8) :incompatible_relds (b) :outer_relds ((b 6) :right_relds (b 8)) :orclause <> :rinfo_serial 13 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selc 0.999469733582758 :outer_select 1 :mergeofamilies (<) :left_em <> :right_em <> :router_is_left false :hashjoinoperator 0 :left_bucketsize 1 :right_bucketsize 1 :left_mvfreq 1 :right_mvfreq 1 :left_hashselector %6 :right_hashselector %6);| RESTRICTINFO :clause (OPEXPR :opnpo 96 :opsresultype 16 :opretset false :opolcid 0 :inputcollid 0 :args ((VAR :varno 6 :varattino 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 6 :varattosyn 1 :location -1}) (VAR :varno 6 :varattino 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 6 :varattosyn 1 :location -1})) :location -1} :is_pushed_down true;can_join true;pseudocountant false;has_clone false;is_clone false;leakproof false;has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 6 9) :required_relds (b 6 9) :incompatible_relds (b) :outer_relds ((b 6) :right_relds (b 9)) :orclause <> :rinfo_serial 26 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selc 0.3333333333333333e-06 :outer_select 0.8171666666666667 :mergeofamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 9 :varattino 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 9 :varattosyn 1 :location -1}) :em_relds ((b 9)) :em_is_const false :em_is_child false :em_data_type 23 :em_jdomain (JOINDOMAIN :jd_relds ((b 1 3 4 6 7 8 9 10))) :right_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 6 :varattino 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 6 :varattosyn 1 :location -1}) :em_relds ((b 6)) :em_is_const false :em_is_child false :em_data_type 23 :em_jdomain (JOINDOMAIN :jd_relds ((b 1 3 4 6 7 8 9 10))) :out_rer_ids ((b 6)) :router_is_left false :hashjoinoperator %6 :left_bucketsize 0.82875670445881e-06 :right_bucketsize 0.479135223326536e-06 :left_mvfreq 0 :right_mvfreq 0 :left_hashselector %6 :right_hashselector %6);| parallel_safe = true | reason = PROPARALLEL_SAFE  
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause (OPEXPR :opnpo 518 :opsresultype 16 :opretset false :opolcid 0 :inputcollid 0 :args ((VAR :varno 6 :varattino 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups
```


[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]


```

Worker 0: actual time=0.444..100.590 rows=253038.00 loops=1
  Buffers: shared hit=3000 read=4500
Worker 1: actual time=0.449..100.752 rows=253539.00 loops=1
  Buffers: shared hit=3095 read=4443
-> Parallel Hash (cost=37034.62..37034.62 rows=394 width=46) (actual time=168.383..168.392 rows=3670.33 loops=3)
  Output: supplier.s_name, supplier.s_suppkey, lineitem.l_orderkey, lineitem.l_partkey, lineitem.l_suppkey, o.o_orderkey
  Buckets: 16384 (originally 1024) Batches: 1 (originally 1) Memory Usage: 1176kB
  Buffers: shared hit=12300 read=15791
Worker 0: actual time=168.382..168.389 rows=3397.00 loops=1
  Buffers: shared hit=3795 read=5847
Worker 1: actual time=168.393..168.397 rows=3803.00 loops=1
  Buffers: shared hit=4638 read=4985
-> Parallel Hash Semi Join (cost=7590.85..37034.62 rows=394 width=46) (actual time=29.986..149.109 rows=3670.33 loops=3)
  Output: supplier.s_name, supplier.s_suppkey, lineitem.l_orderkey, lineitem.l_partkey, lineitem.l_suppkey, o.o_orderkey
  Hash Cond: (lineitem.l_orderkey = o.o_orderkey)
  Buffers: shared hit=12300 read=15791
Worker 0: actual time=29.980..140.693 rows=3397.00 loops=1
  Buffers: shared hit=3795 read=5847
Worker 1: actual time=30.085..152.932 rows=3803.00 loops=1
  Buffers: shared hit=4638 read=4985
-> Hash Join (cost=127.91..29564.73 rows=980 width=42) (actual time=0.771..114.794 rows=9132.33 loops=3)
  Output: supplier.s_name, supplier.s_suppkey, lineitem.l_orderkey, lineitem.l_partkey, lineitem.l_suppkey
  Hash Cond: (lineitem.l_suppkey = supplier.s_suppkey)
  Buffers: shared hit=9100 read=13719
Worker 0: actual time=0.813..106.908 rows=8507.00 loops=1
  Buffers: shared hit=2703 read=4379
Worker 1: actual time=0.786..118.545 rows=9517.00 loops=1
  Buffers: shared hit=3575 read=4285
-> Parallel Seq Scan on public.lineitem (cost=0.00..28801.84 rows=166662 width=12) (actual time=0.246..96.916 rows=252963.00 loops=3)
  Output: lineitem.l_orderkey, lineitem.l_partkey, lineitem.l_suppkey, lineitem.l_linenum, lineitem.l_quantity, lineitem.l_ext
endedprice, lineitem.l_discount, lineitem.l_tax, lineitem.l_returnflag, lineitem.l_shipdate, lineitem.l_receiptdate, lineitem.l_shipinstruct, lineitem.l_shi
pmode, lineitem.l_comment
  Filter: (lineitem.l_receiptdate > lineitem.l_commitdate)
  Rows Removed by Filter: 147027
  Buffers: shared hit=8833 read=13719
Worker 0: actual time=0.256..90.260 rows=235578.00 loops=1
  Buffers: shared hit=2614 read=4379
Worker 1: actual time=0.242..100.157 rows=261595.00 loops=1
  Buffers: shared hit=3486 read=4285
-> Hash (cost=127.76..127.76 rows=12 width=30) (actual time=0.505..0.508 rows=72.00 loops=3)
  Output: supplier.s_name, supplier.s_suppkey
  Buckets: 1024 Batches: 1 Memory Usage: 13kB
  Buffers: shared hit=267
Worker 0: actual time=0.530..0.532 rows=72.00 loops=1
  Buffers: shared hit=89
Worker 1: actual time=0.526..0.527 rows=72.00 loops=1
  Buffers: shared hit=89
-> Hash Join (cost=12.14..127.76 rows=12 width=30) (actual time=0.111..0.495 rows=72.00 loops=3)
  Output: supplier.s_name, supplier.s_suppkey
  Hash Cond: (supplier.s_nationkey = nation.n_nationkey)
  Buffers: shared hit=267
Worker 0: actual time=0.145..0.519 rows=72.00 loops=1
  Buffers: shared hit=89
Worker 1: actual time=0.148..0.515 rows=72.00 loops=1
  Buffers: shared hit=89
-> Seq Scan on public.supplier (cost=0.00..108.00 rows=2000 width=34) (actual time=0.057..0.235 rows=2000.00 loops=3)
  Output: supplier.s_suppkey, supplier.s_name, supplier.s_address, supplier.s_nationkey, supplier.s_phone, supplier.s
_acctbal, supplier.s_comment
  Buffers: shared hit=264
Worker 0: actual time=0.077..0.248 rows=2000.00 loops=1
  Buffers: shared hit=88
Worker 1: actual time=0.081..0.251 rows=2000.00 loops=1
  Buffers: shared hit=88
-> Hash (cost=12.12..12.12 rows=1 width=4) (actual time=0.044..0.044 rows=1.00 loops=3)
  Output: nation.n_nationkey
  Buckets: 1024 Batches: 1 Memory Usage: 9kB
  Buffers: shared hit=3
Worker 0: actual time=0.057..0.057 rows=1.00 loops=1
  Buffers: shared hit=1
Worker 1: actual time=0.057..0.057 rows=1.00 loops=1
  Buffers: shared hit=1
-> Seq Scan on public.nation (cost=0.00..12.12 rows=1 width=4) (actual time=0.037..0.040 rows=1.00 loops=3)
  Output: nation.n_nationkey
  Filter: (nation.n_name = 'ARGENTINA'::bpchar)
  Rows Removed by Filter: 24
  Buffers: shared hit=3
Worker 0: actual time=0.051..0.054 rows=1.00 loops=1
  Buffers: shared hit=1
Worker 1: actual time=0.051..0.054 rows=1.00 loops=1
  Buffers: shared hit=1
-> Parallel Hash (cost=6834.50..6834.50 rows=50275 width=4) (actual time=29.075..29.075 rows=40048.00 loops=3)
  Output: o.o_orderkey
  Buckets: 131072 Batches: 1 Memory Usage: 5760kB
  Buffers: shared hit=3200 read=2072
Worker 0: actual time=29.151..29.152 rows=40621.00 loops=1
  Buffers: shared hit=1092 read=668
Worker 1: actual time=28.925..28.926 rows=39678.00 loops=1
  Buffers: shared hit=1063 read=700
-> Parallel Seq Scan on public.orders o (cost=0.00..6834.50 rows=50275 width=4) (actual time=0.014..23.153 rows=40048.00 loops=3)
  Output: o.o_orderkey
  Filter: (o.o_orderpriority = ANY ('{1-URGENT,2-HIGH}'::bpchar[]))
  Rows Removed by Filter: 59952
  Buffers: shared hit=3200 read=2072
Worker 0: actual time=0.013..23.059 rows=40621.00 loops=1
  Buffers: shared hit=1092 read=668
Worker 1: actual time=0.016..23.318 rows=39678.00 loops=1
  Buffers: shared hit=1063 read=700
-> Parallel Hash (cost=4377.33..4377.33 rows=66660 width=8) (actual time=19.514..19.515 rows=53333.33 loops=3)
  Output: ps.ps_partkey, ps.ps_suppkey
  Buckets: 262144 Batches: 1 Memory Usage: 8352kB
  Buffers: shared read=3544
Worker 0: actual time=19.664..19.666 rows=53560.00 loops=1
  Buffers: shared read=1174
Worker 1: actual time=19.210..19.213 rows=53001.00 loops=1
  Buffers: shared read=1202
-> Parallel Seq Scan on public.partsupp ps (cost=0.00..4377.33 rows=66660 width=8) (actual time=0.082..9.513 rows=53333.33 loops=3)
  Output: ps.ps_partkey, ps.ps_suppkey
  Filter: (ps.ps_availqty > 0)
  Buffers: shared read=3544
Worker 0: actual time=0.087..9.531 rows=53560.00 loops=1
  Buffers: shared read=1174
Worker 1: actual time=0.111..9.584 rows=53001.00 loops=1
  Buffers: shared read=1202
-> Parallel Hash (cost=27551.87..27551.87 rows=499987 width=8) (actual time=120.093..120.094 rows=399989.67 loops=3)
  Output: l2.l_orderkey, l2.l_suppkey
  Buckets: 262144 Batches: 8 Memory Usage: 8032kB
  Buffers: shared hit=8833 read=13719, temp written=3628
Worker 0: actual time=119.010..119.010 rows=395545.00 loops=1
  Buffers: shared hit=3006 read=4431, temp written=1192
Worker 1: actual time=119.009..119.009 rows=395208.00 loops=1
  Buffers: shared hit=2948 read=4478, temp written=1200
-> Parallel Seq Scan on public.lineitem l2 (cost=0.00..27551.87 rows=499987 width=8) (actual time=0.086..48.976 rows=399989.67 loops=3)
  Output: l2.l_orderkey, l2.l_suppkey
  Buffers: shared hit=8833 read=13719
Worker 0: actual time=0.078..48.631 rows=395545.00 loops=1
  Buffers: shared hit=3006 read=4431
Worker 1: actual time=0.099..48.682 rows=395208.00 loops=1
  Buffers: shared hit=2948 read=4478
Planning:
  Buffers: shared hit=227 read=8
Planning Time: 52.156 ms
Execution Time: 517.165 ms
(206 rows)

```

=====

BENCHMARK SUMMARY

```

=====
Runs           : 10
Average Planning : 49.455 ms
Average Execution : 591.556 ms

```

Generated Files:

```

run10_explain.txt
summary.txt
adityadave@Adityas-MacBook-Air-3 Q2&Q21 %

```